

Project: MNCH Commodity Stock Dashboard (Chemonics International)

Background

In Ghana, the availability of Maternal, Newborn, and Child Health (MNCH) commodities and stock-out rates vary across different types of healthcare facilities. According to a supply chain assessment, while stock-out rates for many MNCH commodities are generally low, Oral Rehydration Salts (ORS) have exhibited a relatively higher stock-out rate of 27% at wholesale outlets. Retail pharmacies and wholesalers are less likely to manage injectable products such as gentamicin, magnesium sulfate, and oxytocin. Chemonics International, an international NGO implementing the USAID Global Health Supply Chain project in Ghana, aims to monitor the availability and stock-out rates of selected MNCH commodities—ORS, Zinc Tablets, Amoxicillin Suspension (125 mg/5 ml), Oxytocin Injection (10 mg/ml), Gentamicin (40 mg/ml), Chlorhexidine Digluconate (7.1% Gel), Magnesium Sulfate (50%), and Hydralazine HCl (20 mg/ml Injection)—at some healthcare facilities in the four most underserved regions of Ghana: Northern, North East, Upper East, and Upper West.

The objective of this project was to:

- Track the availability and stock-outs of critical MNCH items.
- Identify trends and supply gaps in the selected healthcare facilities across Northern Ghana.
- Support timely resupply decisions and evidence-based planning.

Approach

1. Data Collection & Sources

- The data was collected weekly from 93 primary healthcare facilities via Kobo Toolbox. Kobo Toolbox data connected to Power BI via Kobo API.

- Selected MNCH Commodities included: *ORS, Zinc Tablets, Amoxicillin Suspension (125 mg/5 ml), Oxytocin Injection (10 mg/ml), Gentamicin (40 mg/ml), Chlorhexidine Digluconate (7.1% Gel), Magnesium Sulfate (50%), and Hydralazine HCl (20 mg/ml Injection).*
- Key fields collected:
 - Facility Name, District, Region
 - Commodity Name
 - Opening Balance, Quantity Received, Quantity Issued, Closing Balance
 - Reported Stock-Out (Yes/No), Days Out of Stock

2. Data Preparation & Modelling

- Cleaned and merged in Power Query.
- Developed a star schema model: Fact table logistics linked to dimension tables: region, product, facility, date_dim.
- Created calculated columns in the date_dim table using DAX (e.g. FirstDateOfWeek, LastDateOfWeek, MonthYear, WeekInMonth, WeekOfDate, etc)
- Created calculated measures using DAX (e.g., Facility_with_MoS, Facility_with_Stockout, SoH, MoS, Last3-12MthsCons etc.)

3. Dashboard Features

- **Visuals:**
 - Matrix showing availability of selected MNCH commodities as well months of stock for each commodity with drill-throughs for district-specific details.
 - Line charts to show trend and compare percentage of facilities running at stockout of selected MNCH commodities on monthly basis.
 - Stacked bar charts to compare percentage of facilities with months of stocks greater than zero for the selected MNCH commodities on monthly basis.

- **Interactivity:**

- **Slicer:** Filter by Year/Month/Weeks, Filter by District/Facility, and Filter by Product.
- **Drill-Throughs:** Applied on Matrix visuals for district-specific details.

4. **Deployment**

- Published to Power BI Service.
- Scheduled data refresh weekly.
- Shared with MNCH program leads and Technical Directors via Power BI service.

Impact

- Enabled real-time monitoring of commodity availability in hard-to-reach regions.
- Facilitated quick redistribution of supplies from overstocked to understocked facilities.
- Informed the Ministry of Health's quarterly procurement and distribution plans.
- Provided evidence to secure additional donor funding for maternal health interventions.